

Ex 1: Two ^{independent.} dice are thrown. Let R.V. X be the sum of the numbers facing up. Find

a) pmf of X

b) $EX \rightarrow$ A) 6 B) 7 C) 12 D) 14

c) $M(t) = M_{g(t)}$

Sol: $X = D_1 + D_2$ $S_x = \{2, 3, \dots, 12\}$
 $1+1 \rightarrow 2$
 $\vdots$
 $6+6 \rightarrow 12$

$$f_X(2) = P(X=2) = P(D_1=1, D_2=1)$$

$$\hookrightarrow \text{pmf of } X = P(D_1=1) \cdot P(D_2=1) = \frac{1}{36}$$

$$f_X(2) = \frac{1}{36}$$

$$f_X(3) = P(X=3) = P(D_1+D_2=3) = \frac{2}{36}$$

$$\begin{array}{l} 1+2 \\ 2+1 \end{array}$$

$$f_X(4) = P(X=4) = P(D_1+D_2=4) = \frac{3}{36}$$

$$\begin{array}{l} 1+3 \rightarrow \frac{1}{36} \\ 2+2 \rightarrow \frac{1}{36} \\ 3+1 \rightarrow \frac{1}{36} \end{array}$$

$$f_X(5) = \frac{4}{36}$$

$$f_X(7) = \frac{6}{36}$$

$$(1+6, 2+5, \dots, 6+1)$$

$$f_X(6) = \frac{5}{36}$$

$$f_X(8) = \frac{5}{36}$$

$$(2+6, 3+5, 4+4, 5+3, 6+2)$$

$$f_X(12) = \frac{1}{36}$$

$f_X(x)$

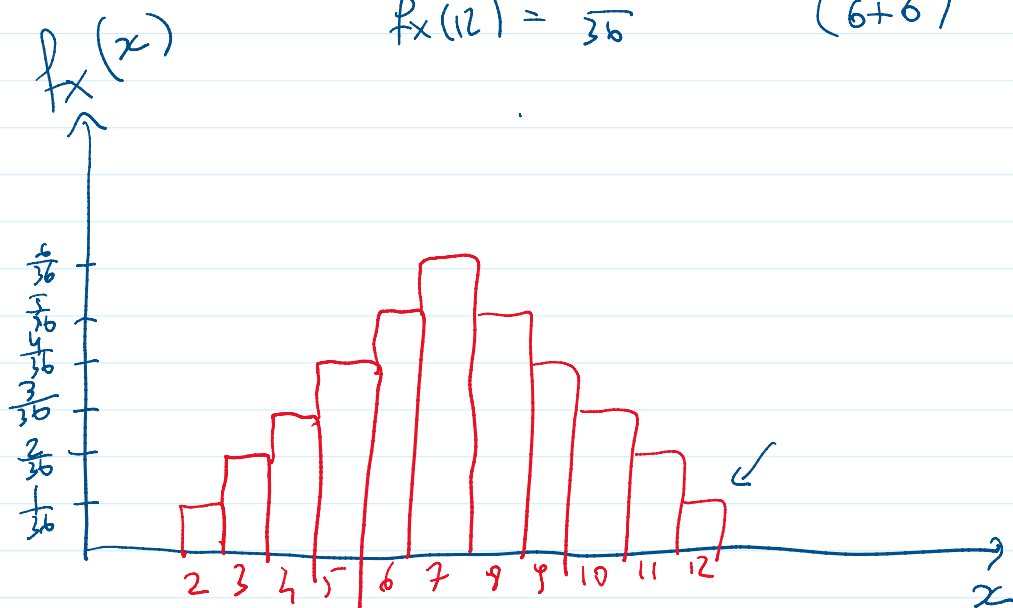
$f_X(x)$

$$f_X(9) = \frac{4}{36}$$

$$f_X(12) = \frac{1}{36}$$

(6+6)

$X = RV$
 x variable.



$$(b) EX = \sum_{x \in \mathcal{X}} x \cdot f_X(x) = \sum_{x=2}^{12} x \cdot f_X(x) =$$

$$= 2 \cdot f_X(2) + 3 \cdot f_X(3) + 4 \cdot f_X(4) + \dots + 12 \cdot f_X(12)$$

$$= 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + 4 \cdot \frac{3}{36} + \dots + 7 \cdot \frac{6}{36} + 8 \cdot \frac{5}{36} + \dots + 12 \cdot \frac{1}{36}$$

$$= \frac{252}{36} = 7$$

Sol 2: $E(X+Y) = EX + EY$

In our problem, $X = D_1 + D_2$

$$EX = ED_1 + ED_2 = 7$$

$$ED_1 = 5 + 0 + 1 + \dots + 1 + \dots + 6 + \dots = \frac{1+12+6}{2}$$

$$ED_1 = \sum_{x \in S_{D_1}} x p_{D_1}(x) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \dots + 6 \cdot \frac{1}{6} = \frac{1+2+\dots+6}{6}$$

$\downarrow$ prob of $D_1=1$ $\downarrow$ prob $D_2=2$ $\downarrow$ prob of $D_2=6$

$$\rightarrow \frac{21}{6} = \frac{7}{2}$$

(c) $M(t) = M_{gf}(t) = M_{\underline{X}}(t)$

$$M(t) = E(\underline{e^{tX}}) = \sum_{x \in S_X} e^{xt} p_X(x) =$$

$$= \sum_{x=2}^{12} e^{xt} p_X(x)$$

$$= e^{2t} \cdot \frac{1}{36} + e^{3t} \cdot \frac{2}{36} + \dots + e^{7t} \cdot \frac{6}{36} + e^{8t} \cdot \frac{5}{36} + \dots + e^{12t} \cdot \frac{1}{36}$$

$\downarrow$ $P(X=2)$ $\downarrow$ $P(X=3)$ $\downarrow$ $P(X=7)$

$$M(0) = E(e^{0X}) = E(1) = 1 \rightarrow \text{property of M.G.T.}$$

$M(t)$ is functn.

$$EX = M'(0)$$

$$EX^2 = M''(0)$$

Ex (2.1.3) For each of the following, determine the constant c so that $f(x)$ satisfies of being a pmf of some RV X

a) $f(x) = x/c$ $x = 1, 2, 3, 4$. A) 1 B) 3 C) 4 D) 6 E) 10

b) $f(x) = c \cdot \frac{1}{4^x}$ $x = 1, 2, 3, \dots$ A) 3 B) $\frac{1}{3}$ C) 4 D) $\frac{1}{4}$ E) 1

Sol: pmf must satisfy 2 conditions (i) $f(x) \geq 0$.

(ii) $\sum_{x \in S_X} f(x) = 1$.

(a). $\sum_{x \in S_X} f(x) = \sum_{x=1}^4 \frac{x}{c} = \frac{1}{c} + \frac{2}{c} + \frac{3}{c} + \frac{4}{c} = \frac{10}{c} = 1 \Rightarrow c = 10$.

(b) $\sum_{x \in S_X} f(x) = \sum_{x=1}^{\infty} c \cdot \frac{1}{4^x} = 1$.

$\downarrow$
 $c \sum_{x=1}^{\infty} \frac{1}{4^x} = c \left(\frac{1}{4} + \frac{1}{4^2} + \dots \right) = \frac{c}{3} = 1$
 geometric series $\Rightarrow c = 3$

Lemma: $a, ar, ar^2, \dots$
 from 3.2A.

$a + ar + ar^2 + ar^3 + \dots = \frac{a}{1-r}$ when $|r| < 1$.

$\frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$

(Ex 2.2.5) let the RV X be the number of days that a certain patient needs to stay in hospital. Suppose X has pmf

$$f(x) = \frac{5-x}{10} \quad x=1, 2, 3, 4$$

If the patient is to receive \$200 from insurance company for each first 2 days of her visit in hospital, and \$100 per day after 2 days what is the expected value of payment will be made to the patient from the Insurance company?

A) 260 B) 300 C) 310 D) 350 E) 360

Sol: $E(\text{Payment})$

$$P(x) = \begin{cases} 200 & x=1 \\ 400 & x=2 \\ 500 & x=3 \\ 600 & x=4 \end{cases}$$

$$E(P(x)) = \sum_{x \in S_X} P(x) \cdot f(x)$$

$$f(x) = \frac{5-x}{10} \quad x=1, 2, 3, 4$$

$$= P(1) \cdot f(1) + P(2) \cdot f(2) + P(3) \cdot f(3) + P(4) \cdot f(4)$$

$$= 200 \cdot \frac{4}{10} + 400 \cdot \frac{3}{10} + 500 \cdot \frac{2}{10} + 600 \cdot \frac{1}{10}$$

$$= 80 + 120 + 100 + 60$$

$$= 360$$